MILITARY SPECIFICATION

PROMETHAZINE HYDROCHLORIDE, IPECAC FLUIDEXTRACT AND POTASSIUM GUAIACOLSULFONATE SYRUP

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Promethazine Hydrochloride, Ipecac Fluidextract and Potassium Guaiacolsulfate Syrup (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA).

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a palatable syrup containing in each 5 ml:

Promethazine Hydrochloride	5 mg
Fluidextract Ipecac	0.01 mg
Potassium Guaiacolsulfonate	44 mg
Citric Acid, Anhydrous	60 mg
Sodium Citrate	197 mg

3.1.1 Alcohol. Shall contain 7 percent alcohol by volume.

3.2 Additives. The coloring, flavoring, sweetening agents and preservatives used in the syrup shall be in the amounts approved by the Federal Food and Drug Administration.

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3.3 Finished product.

3.3.1 Assay.

3.3.1.1 Promethazine hydrochloride. The syrup shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of promethazine hydrochloride when determined as specified in 4.5.1.

3.3.1.2 Alcohol content. The finished preparation shall assay to contain not less than 6.3 percent and not more than 7.7 percent of alcohol by volume when determined as specified in 4.5.3.

3.3.1.3 Potassium guaiacolsulfonate. The finished preparation shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of potassium guaiacolsulfonate when determined as specified in 4.5.4.

3.3.1.4 Citric acid, anhydrous, fluidextract ipecac, and sodium citrate. The amounts of citric acid, anhydrous, fluidextract ipecac, and sodium citrate in the syrup shall be determined from the batch formulation record kept on file by the manufacturer. The content of each of the three ingredients shall be in accordance with the requirements specified in 3.1.

3.3.2 Specific gravity. The specific gravity of the preparation shall be not less than 1.28 and not more than 1.30 at 25° C , when determined, using the USP method.

3.3.3 pH. The pH of the undiluted preparation shall be not less than 4.70 and not more than 5.20 at 25° C, when determined potentiometrically, using the USP method.

3.3.4 Identification. The promethazine in the preparation shall be identified by its ultraviolet absorption spectrum between 380 nm and 280 nm and by yellow crystal formation when determined as specified in 4.5.2.

3.3.5 Flavor and palatability. The finished syrup shall be palatable and pleasant to the taste with no unpleasant after-taste. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) individually packaged samples (each containing 4 fl oz) of the finished syrup, representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of palatability (see 4.6 - Palatability test). The remaining samples will be used by cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for

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use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determined that such product was previously procured and/or tested by the purchasing activity and found to comply with the palatability requirement.

3.4 Ingredients.

3.4.1 Ipecac fluidextract. The ipecac fluidextract entering into the manufacture of the finished preparation shall be in accordance with the tests, standards, and requirements of the USP XVI, which was official from 1 October 1960 to 31 August 1965.

3.4.2 Potassium guaiacolsulfonate. The potassium guaiacolsulfonate entering into the manufacture of the finished preparation shall be in accordance with the tests, standards, and requirements of the NF XIII, which was official from 1 September 1970 to 30 June 1975.

3.5 Other ingredients. All other ingredients entering into the manufacture of the finished preparation shal be of USP or NF quality, or if not included in either of these compendia, the ingredients shall be suitable for use in the preparation.

3.6 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the syrup to the date of delivery to the Government.

3.7 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single uniform, and homogeneous quantity of syrup produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Criteria for sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristics	Requirement paragraph	Inspection level 1/	Minimum sample
Assay:			
Promethazine hydrochloride	3.3.1.1	S-2	5 } Composite
Alcohol content	3.3.1.2	S-2	5 } Composite
Potassium guaiacolsulfonate	3.3.1.3	S-2	5 } Composite

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TABLE II. For end item testing (cont'd)

Characteristics	Requirement paragraph	Inspection level 1/	Minimum sample
specific gravity	3.3.2	S-2	5 } Composite
pH	3.3.3	S-2	5 } Composite
Identification	3.3.4	S-2	5 } Composite
Palatability	3.3.5	-	-

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished syrup shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Syrup not free from foreign matter.
102	Syrup has crystal formation.
103	Syrup not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents and foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5.1 Assay for promethazine hydrochloride.

a. Palladium Chloride Solution. Transfer 500 mg of palladium chloride, $PdCl_2$ to a 500 ml beaker, add 5 ml of hydrochloric acid, and warm on a steam bath until solution is complete. Cool, add 300 ml of purified water and 41 grams of sodium acetate, mix to dissolve and adjust the solution's pH to 3.5 with concentrated hydrochloric acid. Dilute the solution to 500 ml with purified water and store in a glass stoppered bottle.

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b. Standard Preparation. Dissolve a suitable quantity of USP Promethazine Hydrochloride Reference Standard, accurately weighed, in purified water to obtain a solution having a known final concentration of about 1 mg per ml (Protect the solution from light).

c. Procedure. Transfer 2.0 ml of the sample, equivalent to about 2 mg of promethazine hydrochloride, and 2.0 ml of Standard Preparation to separate 125 ml separators containing 10 ml of Palladium Chloride Solution. Shake well for 30 seconds and add 10 ml of a saturated sodium chloride solution to each separator. Extract the palladium-promethazine complex from each one with three 15 ml portions of ethylene dichloride collecting the extracts in 50 ml volumetric flasks through ethylene dichloride-wetted cotton overlaid with anhydrous sodium sulfate (granular). Wash the cotton with ethylene dichloride, using the washings to dilute the extracts to 50.0 ml. Concomitantly measure the absorbances of the extracts at 415 nm with a suitable spectrophotometer in 1 cm cells, and using ethylene dichloride as the blank. Retain the sample extract for Identification Test 4.5.2a., and calculate the assay as follows:

$$\frac{Au}{As} \times \frac{100\ C}{} = \text{percent of required amount}$$

where,

Au = sample absorbance
As = standard absorbance
C = concentration of Standard Preparation, mg/ml

4.5.2 Identification for promethazine hydrochloride.

a. Dilute the sample extract retained from test 4.5.1 with an equal volume of ethylene dichloride. Record its ultraviolet absorption spectrum between 380 and 280 nm, with a suitable recording spectrophotometer, in 1 cm cells, and using ethylene dichloride as the blank. The spectrum exhibits a minimum at 350 $\pm$ 2 nm and a maximum at 305 $\pm$ 2 nm.

b. Transfer 50 ml of sample to a 250 ml separatory funnel, add 50 ml of chloroform and 10 ml of anhydrous methanol. Stopper and shake vigorously for 5 minutes. Allow the two phases to separate and transfer the bottom layer to a 200 ml beaker. Evaporate to dryness on a steam bath with the aid of a current of air. Add 5 ml of 3.5 percent anhydrous methanol picric acid solution and heat on a steam bath for 5 minutes. Cool in an ice bath until yellow crystals appear.

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4.5.3 Alcohol content. Adjust the temperature of the sample to 25° C and transfer exactly 100 ml to a distilling flask. Dilute with 150 ml purified water and distill, collecting about 90 ml distillate in a 100 ml volumetric flask immersed in an ice bath. Rinse the distillate into a separator, add 10 grams NaCl and 30 ml petroleum ether. Shake the mixture to extract volatile oils and $CHCl_3$. Draw off the water-alcohol layer into a distilling flask. Wash the petroleum ether solution in the separator with two 40 ml portions of saturated NaCl solution, adding the aqueous washings to the distilling flask. Distill, collecting about 90 ml distillate in a 100 ml volumetric flask immersed in an ice bath. Adjust the temperature of the distillate to 25° C and dilute to the mark with purified water at the same temperature. Add about 5 gram talc, mix, and filter. Determine the specific gravity of the filtrate at 25°C in a pycnometer. Calculate percent required amount of alcohol. As an alternate, the alcohol content may be determined by a suitable gas chromatography method yielding accurate and reproducible results.

4.5.4 Assay for potassium guaiacolsulfonate.

a. Reagents and Solutions:

1. 4-Aminoantipyrene Hydrochloride Solution: Prepare a 1 percent solution of the reagent in purified water. Store the solution in amber glassware and use within one week.

2. Potassium Ferricyanide Solution: Prepare a fresh 4 percent solution in purified water. (protect from light)

3. Buffer Solution: Dissolve 121.1 grams of tris (hydroxymethyl) aminomethane in about 900 ml of purified water. Adjust the pH to 9.0 with hydrochloric acid and dilute to 1000 ml with purified water.

b. Standard Preparation:

Accurately weigh a suitable quantity of potassium guaiacolsulfonate reference standard and dilute with purified water to obtain a known final concentration of approximately 0.088 mg per ml.

c. Assay Preparation:

Dilute 10.0 ml of the sample (equivalent to 88 mg of potassium guaiacolsulfonate hemihydrate) to 1000.0 ml with purified water.

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d. Procedure:

Into three separate 50 ml volumetric flasks, pipet 3.0 ml portions of Standard Preparation, Assay Preparation, and purified water (blank). To each flask, add 10.0 ml of Buffer Solution, 15.0 ml isopropyl alcohol, and 2.0 ml of Potassium Ferricyanide Solution. Mix well and then add 5.0 ml of 4-Aminoantipyrene Hydrochloride Solution to each flask; mix and dilute to volume with purified water. After 30 minutes, concomitantly measure the absorbances of the sample and standard at 500 nm, with a suitable spectrophotometer, and using the blank in the reference cell. Calculate as follows:

$$\frac{Au}{As} \times \frac{1000C}{88} \times 100 = \text{percent of required amount}$$

where,

Au = sample absorbance
As = standard absorbance
C = concentration of Standard Preparation, mg of potassium guaiacolsulfonate hemihydrate per ml.

4.6 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than Palatability Standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the sample shall be equal to or greater than the average rating of the standard similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ Palatability Standard is available, upon separate request to the contracting officer, Defense Personnel Support Center.

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4.7 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.8 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The equirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.2) and as specified herein.

5.1.1 Unit of Issue. One bottle containing four fl oz (118 ml), as specified, constitutes one unit of issue.

5.2 Packaging. Packaging shall be level A or C, as specifed (see 6.1).

5.2.1 Level A.

5.2.1.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. The container shall be provided with a child-proof safety closure.

5.2.1.2 Unit package. At the option of the contractor, a unit package may be furnished for each 4 fl oz bottle. If a unit package is furnished each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.1.3 Intermediate package. Twelve units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Four intermediate packages (48 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

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5.3.2 Level B. Four intermediate packages (48 units) shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° -30°C (59° -86°F)".

5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6505-01-029-8075	PROMETHAZINE HYDROCHLORIDE, IPECAC FLUIDEXTRACT, AND POTASSIUM GUAIACOLSULFONATE SYRUP, 4 fl oz (118 ml)

6.3 All references of PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1853

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DOCUMENT IDENTIFIER AND TITLE MIL-P-37714 (DSA-DM) PROMETHAZINE HYDROCHLORIDE, IPECAC FLUIDEXTRACT, AND POTASSIUM GUAIACOLSULFONATE SYRUP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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